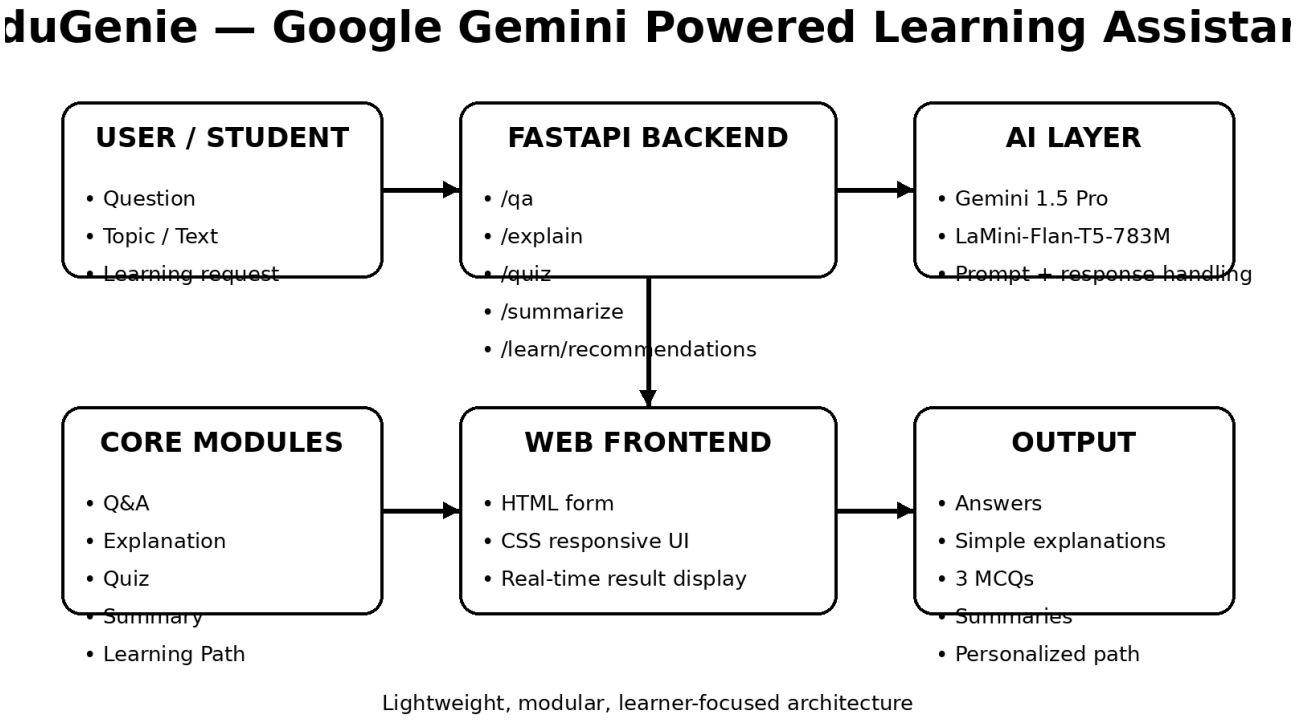
EDUGENIE

Google Gemini Powered Learning Assistant

Project Report



An AI-powered educational assistant for question answering, concept explanation, quiz generation, summarization, and personalized learning recommendations.

Prepared from the supplied EduGenie project document and development reference.

1. Project Overview

EduGenie is a lightweight AI-powered educational assistant designed to simplify learning through generative AI. It supports students and self-learners with concise answers, simplified explanations, quizzes, summaries, and structured learning recommendations.

The supplied project document describes a FastAPI backend with a simple HTML/CSS frontend and AI services based on Google Gemini and a lightweight local model.

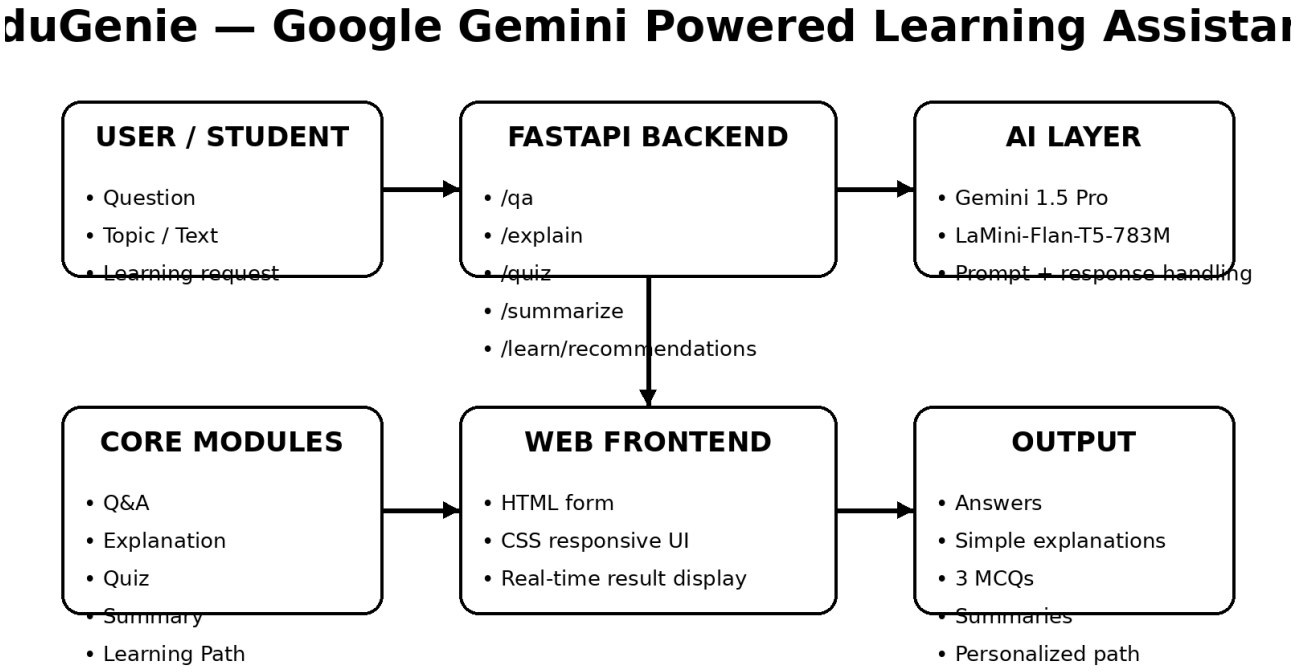
2. Objectives

- Answer academic and general-knowledge questions using AI.
- Explain complex concepts in simple, readable language.
- Generate multiple-choice quizzes from topics or passages.
- Summarize long educational content for quick revision.
- Provide personalized learning paths from beginner to advanced levels.
- Keep the application lightweight and accessible on modest hardware.

3. Technology Stack

Layer	Technology / Component
Backend	Python, FastAPI, Uvicorn
Frontend	HTML, CSS, Jinja2 templates
Cloud AI	Google Gemini 1.5 Pro via API
Local AI	LaMini-Flan-T5-783M for concept explanation
API Style	RESTful POST endpoints
Runtime	Local development with Uvicorn

4. System Architecture



Lightweight, modular, learner-focused architecture

5. Core Functional Modules

Explanation Module

Uses LaMini-Flan-T5-783M to produce concise, context-aware explanations that break complex concepts into understandable language.

Q&A; Module

Uses Gemini 1.5 Pro for general knowledge and academic question-answering with contextual AI assistance.

Quiz Module

Generates three multiple-choice questions from a supplied passage, with four options for each question. The project specifies structured JSON output and response cleaning before parsing.

Summary Module

Condenses long paragraphs into clear summaries while retaining core information and removing redundancy.

Learning Path Module

Generates a structured learning path from beginner to advanced concepts, with suggested resources and guidance adapted to the learner’s level.

6. Backend API and Workflow

RESTful endpoints defined in the supplied project:

- POST /qa
- POST /explain
- POST /quiz
- POST /summarize
- POST /learn/recommendations

Typical request flow

User input → FastAPI endpoint → module logic → AI model/API → response processing → result displayed in the web interface.

The development reference also emphasizes validation, modular AI services, health checking, API documentation, and separation between frontend and backend responsibilities.

7. Frontend

- Task selection: Explain, QnA, Quiz, Summary, or Recommend Path.
- Text area for learner input.
- Submit action connected to the FastAPI backend.
- Responsive CSS styling.
- Generated result displayed below the input interface.

8. Installation and Local Execution

Prerequisites

- Python 3.10 or newer.
- FastAPI and Uvicorn environment.
- Jinja2 for HTML templating.
- Google Gemini API key for cloud AI functions.

Basic setup

1. Install Python 3.10+ and verify with **python --version**.
2. Install dependencies from **requirements.txt**.
3. Configure the Google Gemini API key securely.
4. Start with **uvicorn main:app --reload**.
5. Open **http://127.0.0.1:8000**.

9. Functional Testing

- Ask a question and verify the response.
- Request an explanation of a concept.
- Generate a three-question quiz with four options per question.
- Summarize a long educational paragraph.
- Request a personalized learning path.

10. Conclusion

EduGenie presents a modular AI learning assistant that combines cloud-based generative intelligence with a lightweight web interface. Its main functions cover question answering, simplified explanations, quiz generation, summarization, and personalized learning recommendations.

The project document identifies challenges including AI-response accuracy, device performance, prompt tuning, user experience, and data privacy. It also describes learning outcomes in prompt engineering, cloud integration, user-centered design, and scalable architecture.

11. Future Scope

- Voice-based learning and spoken interaction.
- Multilingual support.
- Mobile application and offline capabilities.
- Progress dashboards and learning analytics.
- Gamification through badges, streaks, and challenges.
- Adaptive learning paths.
- Group study, teacher/parent dashboards, and LMS integration.
- Image/PDF input for doubt solving and summarization.
- Smart notifications and real-time synchronization.

Source note

This report is based primarily on the supplied EduGenie project document. The supplied “5. Project Development” PDF was used only as a reference for organizing implementation, workflow, API, frontend, and testing sections; EduGenie-specific details were retained from its own document.